Inference-Time Diversity in RL-Trained Lean Theorem Provers

A Diagnostic Study

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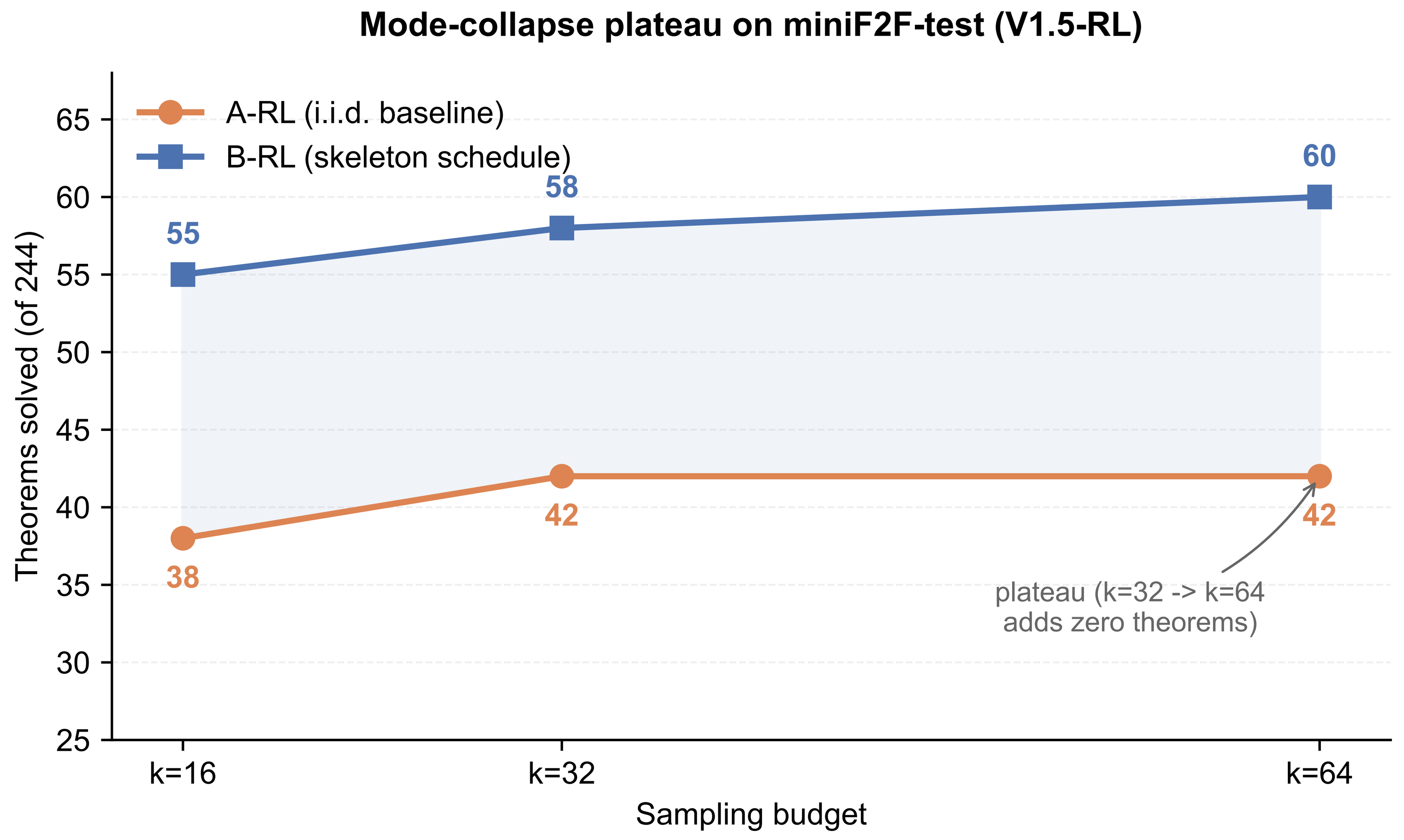
TL;DR *RL-trained Lean theorem provers run out of ideas as you sample more. A fixed schedule of 15 tactic skeletons closes the gap, and the effect is RL-specific.*

Setup

Models (all 7B): primary within-architecture analysis on **DeepSeek-Prover-V1.5-RL** paired with non-RL **V1.5-Base**; cross-model replication on RL-trained **DeepSeek-Prover-V2-7B** (reasoning-mode) and SFT-trained **Goedel-Prover**.
Benchmark: miniF2F-test (244 Lean 4 / Mathlib theorems).
Schedule: 15 hand-picked tactic skeletons (`simp`, `intro`, `constructor`, `aesop`, ...) $\times$ 8 NL goal hints = 120-slot grid. Budgets $k \in \{16, 32, 64\}$ take the first k slots.
Decoding: temperature 0.6, top-p 0.95; max 1024 tokens (V1.5 / Goedel completion-mode) or 8192 tokens (V2 reasoning-mode). **Verifier:** `lake env lean --json`, 120 s timeout, 8-way sharding by $i \bmod 8$.

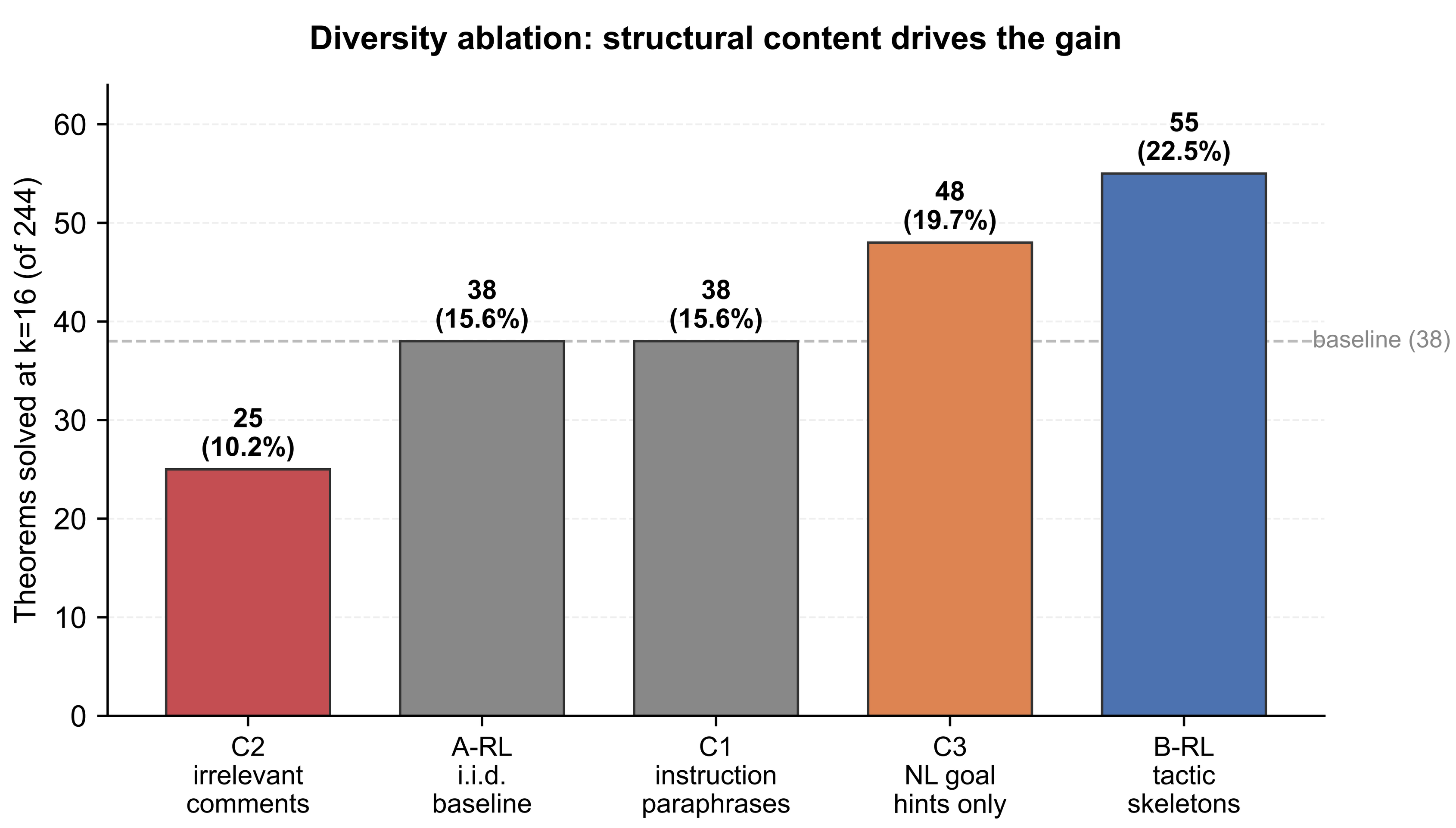
Mode collapse: A-RL plateaus at $k{=}32$



	$k{=}16$	$k{=}32$	$k{=}64$
A-RL (i.i.d.)	38/244 (15.6%)	42/244 (17.2%)	42/244 (17.2%)
B-RL (skeleton)	**55** (22.5%)	**58** (23.8%)	**60** (24.6%)
Δ absolute	$+17$	$+16$	$+18$
Δ relative	$+44.7\%$	$+38.1\%$	$+42.9\%$

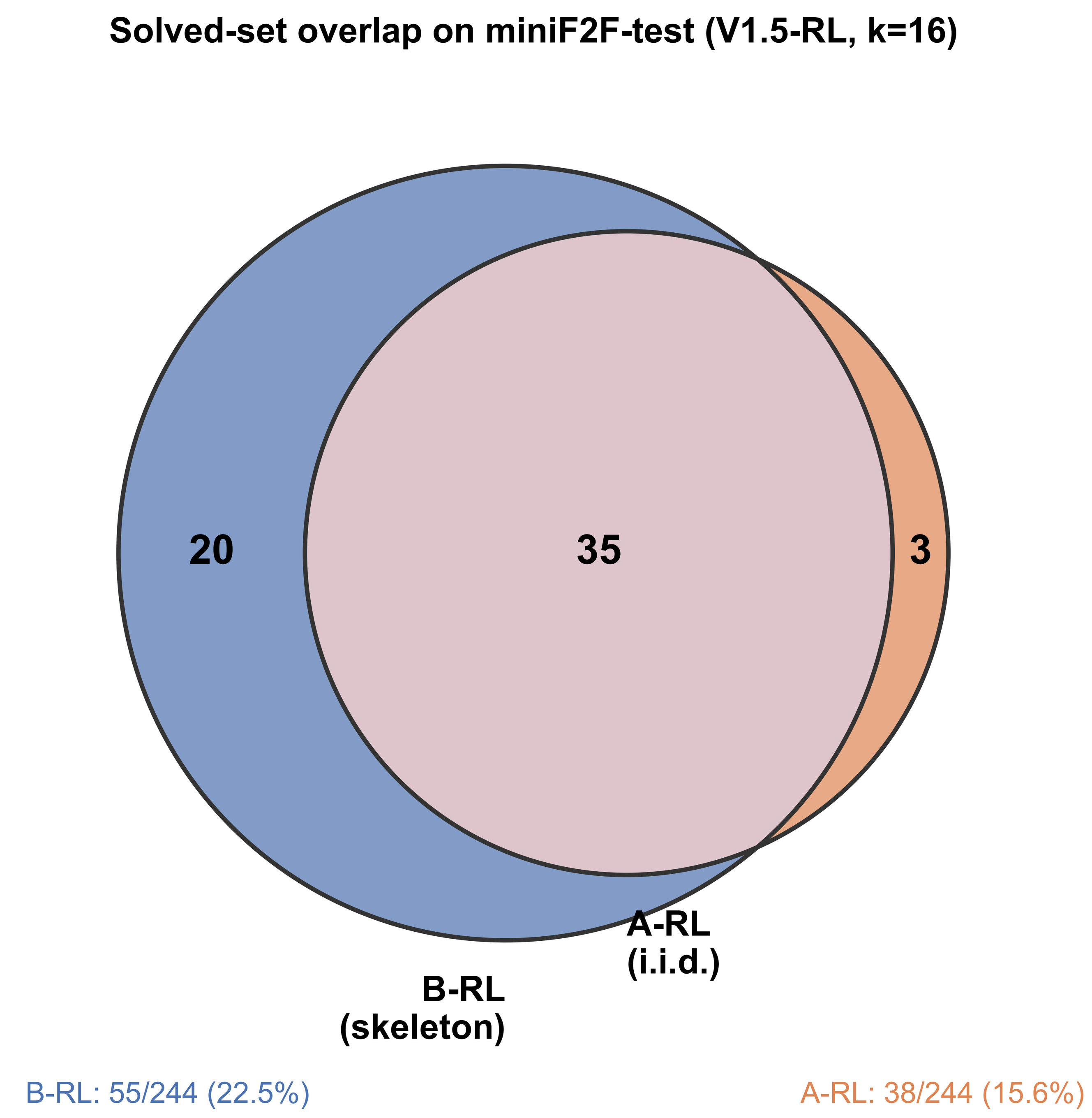
Across-seed: mean $\Delta = +12.3 \pm 4.2$ theorems at $k{=}16$ ($n{=}3$ seeds, sign positive every seed).

Diversity ablation: where the gain comes from



The A$\to$B gap at k=16 is 17 theorems. C3 (NL hint, no tactic prefix) recovers 10 of those 17; the tactic stem (B) picks up the other 7. Paraphrased instructions (C1) add nothing; irrelevant comments (C2) drop solves by 13. The model already has the proofs; what changes between A and B is which head of a peaked sampling distribution gets reached.

Solved-set overlap (V1.5-RL, $k{=}16$)



20 theorems are solved only by B-RL, **3** only by A-RL, **35** by both.

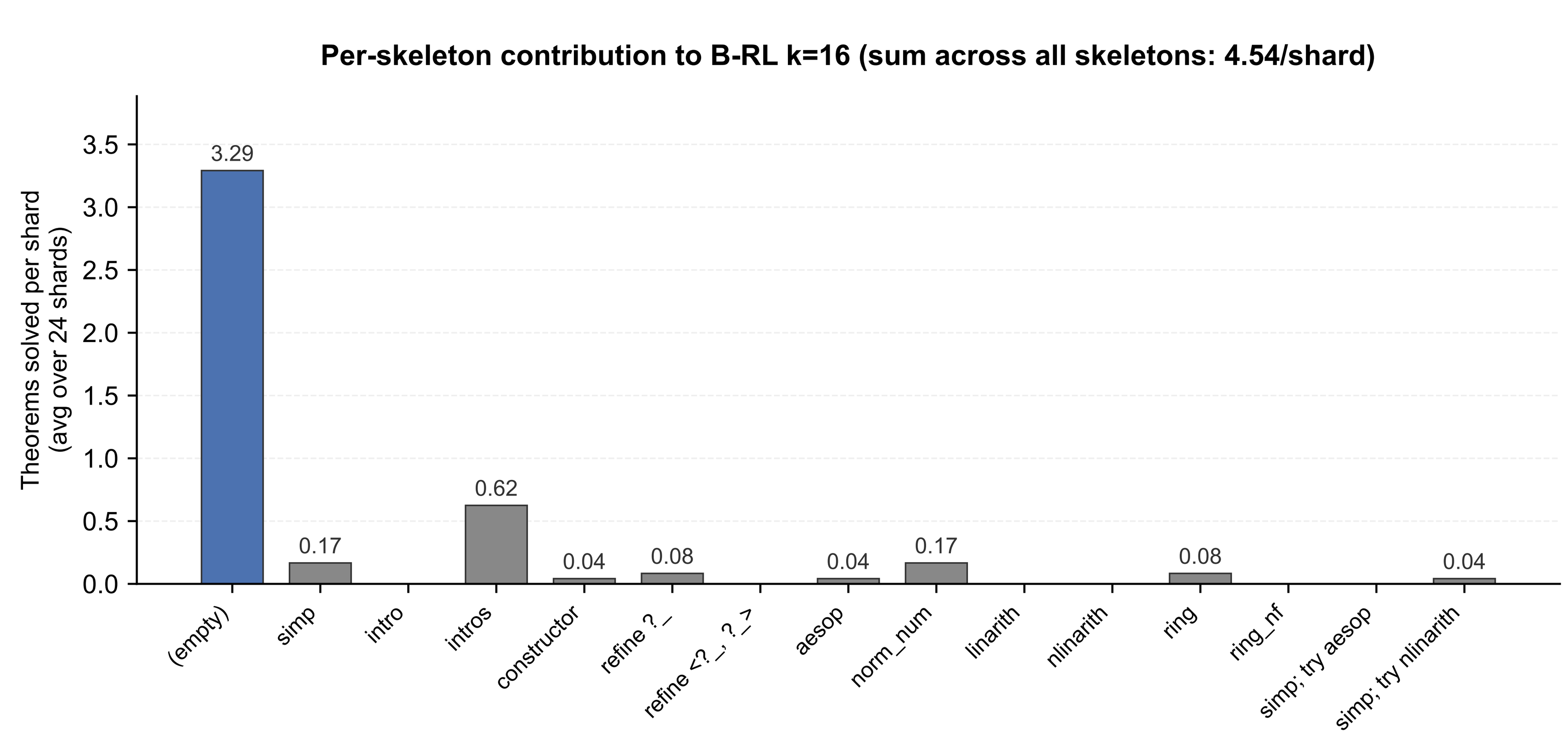
Phenomenon is RL-specific

V1.5-Base (same model, no RL fine-tune) proves **0/244** theorems at every k, with or without skeletons.

- Base outputs `sorry` or echoes the statement in 58.8% of attempts.
- RL model emits `sorry` in $<0.2\%$ of attempts. Its failures are valid but wrong `rw`/`apply` sequences.

RL produces both the proof ability and the narrow sampling that caps it.

Per-skeleton contribution (within $k{=}16$)



24 B-RL $k{=}16$ shards, within-shard counting. The `(empty)` bar is the C3 condition embedded inside B-RL: no tactic prefix, cycling through the 8 hint variants. It accounts for 3.29 of 4.54 solves per shard. The other 14 skeletons split the remaining 1.25, which closes A-RL's gap to B-RL.

Cross-model generalisation

A$\to$B on two more 7B Lean provers:

- RL-trained **V2-7B**: $+3$ frontier theorems no sampling baseline reaches.
- SFT-trained **Goedel-Prover**: -10.0 ± 4.4 theorems across $n{=}3$ seeds. Skeletons hurt SFT.

Distributional evidence of collapse

Across **13,159** V1.5-RL samples, the median theorem produces only **2** distinct first tactics. For **49%** of theorems the model picks the same first tactic every time.

Code, prompts, per-attempt logs: `github.com/zacn04/icml_hints_for_provers`